

NANDAN SRIRANGA

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SUMMARY

PhD candidate in Electrical Engineering with experience designing, implementing, debugging, and evaluating computational research projects in statistical signal processing, sequential inference, distributed detection, sensor fusion, and machine learning. Strong background in building MATLAB/Python simulation pipelines, running Monte Carlo experiments, using MATLAB Parallel Computing Toolbox and `parfor` loops for large-scale simulations, analyzing algorithm failure modes, and converting mathematical ideas into reproducible computational results. Interested in applied research roles involving computational experimentation, ML system evaluation, document understanding, signal processing, and first-principles debugging of complex technical systems.

- Author of 12 accepted papers in peer-reviewed journals and conferences, with 3 additional journal manuscripts under review or in preparation.
- Served as a reviewer for IEEE Transactions on Aerospace and Electronic Systems, IEEE Transactions on Signal Processing, IEEE Communications Letters, IEEE Information Theory Workshop, and EUSIPCO.
- Used Mathpix during my PhD workflow to convert equations into LaTeX code, giving me first-hand appreciation for the document-understanding problem the company is solving.
- Experienced in presenting research at conferences and academic research events.

CORE STRENGTHS

- **Computational research:** Designed multi-month research projects from mathematical formulation to simulation, debugging, evaluation, and publication.
- **Large-scale simulation:** Built MATLAB/Python pipelines for Monte Carlo experiments, ROC analysis, detection-delay studies, and accuracy/complexity tradeoff evaluation.
- **Parallel MATLAB workflows:** Used MATLAB Parallel Computing Toolbox and `parfor` loops to accelerate repeated simulation trials and parameter sweeps.
- **Statistical inference:** Developed detection, estimation, sequential testing, anomaly detection, and adaptive sensing algorithms for noisy and uncertain data.
- **Self-directed learning:** Learned the technical tools required by each research problem, including statistical learning, CNNs, SVMs, MCTS, universal hypothesis testing, and martingale-based analysis.

TECHNICAL SKILLS

Programming & Simulation: MATLAB, Python, C++, MATLAB Parallel Computing Toolbox, `parfor`, Monte Carlo simulation, algorithm debugging, performance evaluation, data visualization

Machine Learning & Data Analysis: CNNs, SVMs, PCA, feature extraction, classification, regression, model evaluation, accuracy/complexity tradeoffs, Monte Carlo Tree Search

Signal Processing & Inference: Detection Theory, Estimation Theory, Sequential Detection, Universal Hypothesis Testing, Anomaly Detection, Sensor Fusion, Distributed Detection, Passive Radar

Mathematics: Probability, Statistics, Random Processes, Linear Algebra, Numerical Linear Algebra, Optimization, Information Theory, Statistical Modeling

EDUCATION

PhD Candidate in Electrical Engineering, Syracuse University

2019–Present

GPA: 3.83/4.0 (52 credits)

Expected Graduation Date: October 2026

B.E. Electronics and Communications Engineering, PES University, Bangalore, India

2014–2018

GPA: 9.43/10

SELECTED RESEARCH SNAPSHOT

- **Sequential universal two-sample testing:** Developed adaptive and active-sensing tests for comparing two unknown data-generating distributions without parametric likelihood models.
- **Moving-target detection:** Built simulation pipelines for adaptive sensor collaboration and GLRT-based target detection using distributed sensor measurements.
- **Network anomaly detection:** Designed low-delay quickest detection methods for identifying link outages and sparse heterogeneous failures in sensor networks.
- **Passive radar sensing:** Developed signal-processing methods using whitening, subspace transformations, cross-correlation, and receiver collaboration.
- **Machine learning projects:** Built CNN/MCTS and SVM/PCA pipelines for decision-making and image-classification tasks.

COMPUTATIONAL RESEARCH EXPERIENCE

PhD Researcher, Syracuse University

Advisor: Prof. Pramod K. Varshney

Sequential Universal Two-Sample Testing with Adaptive and Active Sensing

- Developed sequential universal two-sample tests for comparing two unknown data-generating distributions without assuming fully specified parametric likelihood models.
- Designed adaptive tests that assign larger weights to more informative data streams while achieving optimal error exponents.
- Developed active stream-selection policies that sample the most informative data source at each round under controlled sensing constraints.
- Used supermartingale theory and convergence arguments to control error probabilities and characterize stopping behavior.
- Built simulation studies to evaluate stopping behavior, error probabilities, adaptive weighting, and active-sensing policies.
- Relevant to online model evaluation, distribution-shift detection, adaptive data collection, and monitoring ML systems under changing data distributions.

Online Sensor Collaboration for Moving Target Detection

- Developed adaptive sensor-collaboration algorithms for detecting a moving target with unknown amplitude, initial position, and velocity.
- Designed a grid-search assisted amplitude plug-in GLRT detector and recursive sensing framework using sequential target-state estimates.
- Built MATLAB simulation pipelines to evaluate detection probability, false-alarm probability, collaboration-noise robustness, and network topology tradeoffs.
- Used repeated Monte Carlo trials and parameter sweeps to compare algorithm performance across sensing, communication, and topology regimes.
- Used MATLAB Parallel Computing Toolbox and `parfor` loops to accelerate computationally expensive simulation experiments.
- Derived asymptotic guarantees showing that the online adaptive design approaches oracle-aided offline collaboration performance.

Quickest Detection of Link Outages and Time-Series Network Anomalies

- Developed low-delay statistical detectors for identifying abrupt failures and anomalies in distributed wireless sensor networks.
- Derived a locally most-powerful statistic and integrated it into a CuSum-style quickest detection framework.
- Reduced computational complexity by avoiding exhaustive evaluation of all possible outage configurations.
- Evaluated detection-delay and false-alarm tradeoffs across outage durations, failure probabilities, sensor subsets, and detection windows.
- Extended the work toward sparsity-aware penalized Rao-score statistics for heterogeneous outage detection using elastic-net, graph fused lasso, and group-lasso penalties.
- Relevant to time-series anomaly detection, predictive maintenance, sensor-network monitoring, and cyber-physical system health diagnostics.

Passive Radar Target Detection with Inter-Receiver Collaboration

- Developed signal-processing methods for distributed two-channel passive radar using reference and surveillance-channel measurements.
- Modeled the unknown illuminator-of-opportunity waveform as lying in a known low-dimensional communication-signal subspace.
- Applied whitening, subspace transformations, and cross-correlation methods to exploit waveform subspace structure.
- Designed inter-receiver collaboration weights so that only a subset of receivers transmit to the fusion center, reducing uplink bandwidth.
- Evaluated ROC performance against no-collaboration and amplify-and-forward baselines.
- Demonstrated improved robustness under weak communication links.

Human-Assisted Decision-Making with Correlated Observations

- Studied binary hypothesis testing where a machine receives a continuous-valued observation and a human provides discrete side information.
- Modeled statistical dependence between human and machine observations using copula-based models.
- Derived likelihood-ratio fusion rules for combining heterogeneous information sources.
- Evaluated detection probability, false-alarm probability, and Bayesian error probability across dependence regimes.
- Relevant to human-assisted AI, heterogeneous information fusion, and trustworthy decision support.

Collaborative Precoding for Covert Communication

- Formulated an optimization problem balancing receiver SNR against eavesdropper detectability in distributed wireless networks.
- Derived a closed-form collaboration matrix design using a generalized Rayleigh-quotient formulation.
- Demonstrated through simulations that collaboration improves receiver reliability while reducing detectability by a passive eavesdropper.
- Connected the work to wireless signal processing, distributed communication systems, and privacy-aware inference.

MACHINE LEARNING PROJECTS

AI Agent for Progressive Chess

- Developed a sequential decision-making agent for progressive chess using Monte Carlo Tree Search with UCT-based exploration.
- Designed a custom heuristic and trained a convolutional neural network to learn state-value priors.
- Integrated CNN outputs into the MCTS framework to guide search toward stronger moves.
- Improved decision efficiency by reducing the effective complexity of a rapidly expanding search space.

Statistical Learning and Feature Design for Image Classification

- Built SVM classifiers with RBF kernels for MNIST digit classification using grid search and 5-fold cross-validation.
- Achieved 99.6% binary classification accuracy and around 93% multi-class test accuracy using one-versus-all classification.
- Designed PCA-based and symmetry-based features that reduced input dimension from 784 to 5 while preserving strong accuracy.
- Analyzed accuracy versus computational-efficiency tradeoffs for high-dimensional and compact feature representations.

RELEVANT COURSEWORK

Signal Processing & Inference: Detection & Estimation Theory, Digital Signal Processing, Random Processes, Digital Communications, Information Theory

Mathematics & Optimization: Probability & Statistics, Linear Algebra, Convex Optimization, Statistical Ranking, Selection & Multiple Comparisons, Linear Models

Machine Learning: Introduction to AI, Big Data, Statistical Learning Theory

TEACHING EXPERIENCE

Teaching Assistant, Syracuse University

- Teaching experience in signal processing and systems courses, including Linear Signals & Systems, Probability & Statistics, and Digital Design Labs.

RESEARCH SERVICE

Peer Reviewer

- Reviewed manuscripts for IEEE Transactions on Aerospace and Electronic Systems, IEEE Transactions on Signal Processing, IEEE Communications Letters, IEEE Information Theory Workshop, and EUSIPCO.
- Evaluated technical papers for mathematical correctness, clarity of presentation, experimental methodology, novelty, and relevance to the signal processing and communications research communities.

PUBLICATIONS

In Preparation/Under Peer Review

1. **Nandan Sriranga**, A. Grootveld, and P. K. Varshney, "Online sensor collaboration for the detection of a moving target in distributed sensor networks," **under review** by IEEE Transactions on Signal Processing, 2026.
2. **Nandan Sriranga** and P. K. Varshney, "Sparsity-aware penalized Rao-score based statistics for quickest detection of heterogeneous link outages," **in preparation** for submission to IEEE Transactions on Signal Processing.

3. **Nandan Sriranga**, A. Grootveld, and P. K. Varshney, “Sequential Universal Two-Sample Testing Using Predictive Likelihoods with Adaptive and Active-Sensing Policies,” **in preparation** for submission to IEEE Transactions on Information Theory.

Accepted Papers

1. **Nandan Sriranga**, H. Yang, and P. K. Varshney, “Quickest detection of link outages in wireless sensor networks using a locally most-powerful statistic,” accepted for presentation at IEEE International Conference on Information Fusion (FUSION), Trondheim, Norway, 2026.
2. **Nandan Sriranga**, H. Yang, V. Gandikota, and P. K. Varshney, “Target detection in two-channel passive radars with inter-receiver collaboration,” in IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), Barcelona, Spain, 2026.
3. A. Grootveld, H. Yang, **Nandan Sriranga**, B. Chen, V. Gandikota, and J. Pollack, “Asymptotically optimal quantum universal quickest change detection,” IEEE International Symposium on Information Theory (ISIT), Guangzhou, China, 2026.
4. **Nandan Sriranga** and P. K. Varshney, “Covert communication in distributed multi-user wireless networks using inter-user collaboration,” 2025 59th Asilomar Conference on Signals, Systems, and Computers, 2025.
5. **Nandan Sriranga**, A. Grootveld, and P. K. Varshney, “Linear sensor collaboration for distributed parameter estimation in the presence of communication failures,” in 2025 28th International Conference on Information Fusion (FUSION), pp. 1–8, IEEE, 2025.
6. C. Quan, S. Bulusu, B. Geng, Y. S. Han, **Nandan Sriranga**, and P. K. Varshney, “On ordered transmission based distributed Gaussian shift-in-mean detection under Byzantine attacks,” IEEE Transactions on Signal Processing, vol. 71, pp. 3343–3356, 2023.
7. C. Quan, **Nandan Sriranga**, H. Yang, Y. S. Han, B. Geng, and P. K. Varshney, “Efficient ordered-transmission based distributed detection under data falsification attacks,” IEEE Signal Processing Letters, vol. 30, pp. 145–149, 2023.
8. **Nandan Sriranga**, A. Trezza, S. Bulusu, D. J. Bucci, and P. K. Varshney, “Online identification of recurring changepoints,” in 2022 56th Asilomar Conference on Signals, Systems, and Computers, pp. 426–430, IEEE, 2022.
9. **Nandan Sriranga**, B. Geng, and P. K. Varshney, “On human assisted decision making for machines using correlated observations,” in 2020 54th Asilomar Conference on Signals, Systems, and Computers, pp. 1502–1506, IEEE, 2020.
10. **Nandan Sriranga**, C. R. Murthy, and V. Aggarwal, “A method to improve consensus averaging using quantized ADMM,” in 2019 IEEE International Symposium on Information Theory (ISIT), pp. 507–511, IEEE, 2019.
11. **Nandan Sriranga**, K. G. Nagananda, and R. S. Blum, “Shared channel ordered transmissions for energy-efficient distributed signal detection,” IEEE Communications Letters, vol. 23, no. 1, pp. 96–99, 2018.
12. **Nandan Sriranga**, K. G. Nagananda, R. S. Blum, A. Saucan, and P. K. Varshney, “Energy-efficient decision fusion for distributed detection in wireless sensor networks,” in 2018 21st International Conference on Information Fusion (FUSION), pp. 1541–1547, IEEE, 2018.